

Day 20 – ATS Production Readiness

1. Overview

The Zecpath AI Applicant Tracking System (ATS) has completed its planned development, integration, testing, optimization, and final validation stages.

The system provides a modular workflow for processing candidate resume information, evaluating candidate-job relevance, generating configurable ATS scores, applying fairness controls, ranking candidates, and producing shortlisting decisions.

Day 20 focuses on reviewing the completed ATS module and validating its readiness for demonstration and further production integration.

2. Core ATS Capabilities

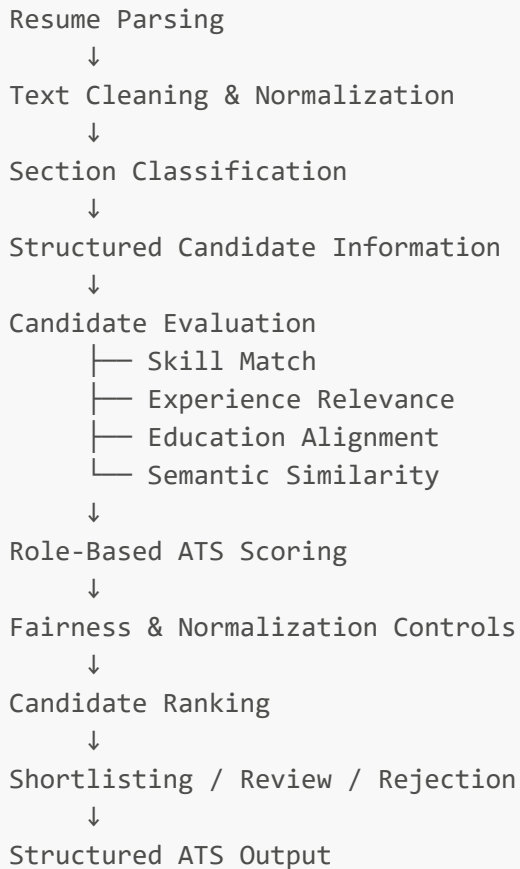
The completed ATS system includes the following major capabilities:

- Resume text extraction and processing
 - Resume text cleaning and normalization
 - Resume section classification
 - Profile/header information handling
 - Structured resume information processing
 - Skill matching
 - Experience relevance evaluation
 - Education relevance evaluation
 - Semantic resume-job matching
 - Configurable role-based ATS scoring
 - Candidate score generation
 - Fairness and scoring normalization controls
 - Candidate ranking
 - Automated shortlisting
 - Review and rejection zones
 - REST API integration
 - Structured API responses
 - Error handling and logging
 - Performance optimization
 - ATS system evaluation
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3. ATS Processing Workflow

The high-level ATS workflow is:

Resume Input
↓



This modular structure allows individual ATS components to be maintained and improved independently.

4. Configurable ATS Scoring

Candidate evaluation uses four major scoring components:

1. Skill Match
2. Experience Relevance
3. Education Alignment
4. Semantic Similarity

The final ATS score is calculated using configurable role-based weights.

```
Overall ATS Score =  
  
(Skill Match × Skill Weight)  
+  
(Experience Relevance × Experience Weight)  
+  
(Education Alignment × Education Weight)  
+  
(Semantic Similarity × Semantic Weight)
```

This allows candidate evaluation priorities to vary according to the target job role while retaining a consistent scoring framework.

5. Fairness and Normalization Controls

Fairness-related controls were incorporated into the ATS evaluation workflow.

Implemented controls include:

- Resume normalization
- Personal attribute masking
- Keyword dependency reduction
- Score normalization
- Semantic contribution monitoring
- Bias indicator evaluation

Testing confirmed that non-essential personal attributes can be masked before candidate evaluation and that fairness indicators can be generated for review.

These controls are intended to reduce unnecessary dependence on personal information and excessive keyword-based scoring.

6. Candidate Ranking and Shortlisting

After ATS scoring, candidates can be sorted according to their overall scores.

The shortlisting module categorizes candidates into:

- **Shortlisted**
- **Review**
- **Rejected**

This provides a recruiter-friendly output while preserving a review zone for candidates who fall between strong acceptance and rejection thresholds.

7. API Integration Readiness

The ATS functionality is accessible through the developed REST API layer.

The API supports major ATS operations including:

- Resume parsing
- Candidate scoring
- Candidate shortlisting
- Request validation
- Structured success responses
- Structured error responses
- Job-processing error handling

Final API integration testing completed successfully.

8. Final System Validation

Before the Day 20 final review, the major ATS components were regression-tested.

The following validation areas completed successfully:

Validation Area	Status
Semantic Matching Engine	PASSED
ATS Scoring Engine	PASSED
Candidate Ranking & Shortlisting	PASSED
Fairness & Normalization	PASSED
ATS API Integration	PASSED
ATS System Evaluation	PASSED
Performance Optimization	PASSED

This confirms that the major ATS modules continued to operate correctly during the final review.

9. ATS Evaluation Metrics

The final ATS system evaluation used 12 controlled test cases covering:

- Tech roles
- Non-tech roles
- Fresher resumes
- Senior profiles

The measured results were:

Metric	Result
Total Test Cases	12
Correct Predictions	11
Incorrect Predictions	1
Accuracy	91.67%
Precision	100.00%
Recall	87.50%
F1 Score	93.33%
Mismatch Cases	1

Tech-role, fresher-resume, and senior-profile test categories achieved 100% accuracy within the controlled test set.

The non-tech category achieved 66.67% accuracy due to one borderline Business Analyst mismatch.

10. Known Evaluation Limitation

One mismatch remains in the controlled ATS evaluation dataset.

A Business Analyst candidate with an ATS score of **69.05** was classified as **REJECT** by the configured decision logic while the manual reference decision was **SELECT**.

This indicates that borderline threshold cases, particularly for non-technical roles, should continue to receive human review and may benefit from future threshold calibration using a larger validated dataset.

The mismatch is documented rather than hidden so that future refinement can be based on measurable evaluation evidence.

11. Performance Validation

Performance optimization testing successfully validated:

- Noisy resume cleaning
- Header/Profile detection
- Resume normalization
- PDF text extraction
- Semantic model reuse
- Semantic batch processing
- Memory stability

The final performance test completed successfully.

Measured execution times vary according to the local system, model state, caching, and runtime environment, so performance should continue to be benchmarked in the intended deployment environment.

12. Day 20 Live Demonstration

A synthetic demonstration dataset was created for a **Data Scientist** role.

The demonstration contained three candidate profiles representing:

- Strong Match
- Moderate Match
- Low Match

The existing ATS scoring, ranking, and shortlisting components were used to produce the final results.

Rank	Candidate	ATS Score	Decision
1	Candidate A	94.60	Shortlisted
2	Candidate B	68.70	Review

Rank	Candidate	ATS Score	Decision
3	Candidate C	25.00	Rejected

The live demonstration completed successfully.

The demo used calculated skill overlap together with controlled synthetic evaluation inputs for experience relevance, education alignment, and semantic similarity. The existing ATS Scoring Engine then calculated the overall score, after which the existing ranking and shortlisting modules generated the final ordering and decisions.

Therefore, the demonstration validates integration of the scoring-to-shortlisting workflow without representing the controlled component inputs as independently predicted values.

13. Production Readiness Assessment

Based on the completed implementation and controlled validation, the ATS module demonstrates readiness for integration-oriented use and further production hardening.

The system currently demonstrates:

- Modular ATS architecture
- Structured resume processing
- Explainable candidate scoring
- Configurable role-based weighting
- Semantic matching capability
- Fairness-related controls
- Automated candidate ranking
- Shortlisting and review decisions
- REST API integration
- Structured error handling
- Successful regression testing
- Performance optimization
- Technical documentation
- Developer documentation
- Architecture documentation

14. Remaining Production Considerations

Before deployment in a real recruitment environment at scale, the following should continue to be considered:

- Validation using larger and more diverse resume datasets
- Role-specific threshold calibration
- Monitoring of false-negative and borderline cases
- Expanded fairness evaluation
- Authentication and authorization for deployed APIs
- Secure resume and candidate-data handling
- Deployment-level monitoring

- Dependency/version maintenance
- Production logging and observability
- Load and concurrency testing
- Human review for consequential hiring decisions

These items represent normal production-hardening activities and opportunities for continued improvement.

15. Final Readiness Status

Day 20 ATS Final Review Status: COMPLETED

The ATS has successfully completed its planned development and controlled validation workflow.

The current implementation can be presented as a completed ATS AI module for the internship deliverable, with documented evaluation results, known limitations, and identified production-hardening considerations.

Conclusion

The completed ATS system integrates resume processing, candidate relevance evaluation, semantic matching, configurable scoring, fairness controls, candidate ranking, automated shortlisting, and API integration into a modular workflow.

Final regression testing and the Day 20 synthetic live demonstration completed successfully.

The system therefore provides a validated foundation for ATS integration and future production hardening while retaining human review as an important part of real-world recruitment decision-making.